Weather Forecasting for Renewable Energy

Requirements by Technology, Department and Horizon, with Evidence from Three Global Models over Germany

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November 2, 2026

Abstract

Wind and solar output, grid operation and electricity prices all depend on weather forecasts, but a forecast means something different to a trader, a commissioning engineer and a lender. This paper maps what a good weather forecast must deliver for each renewable technology, for each department across the asset life cycle (development, financing, engineering, construction, commissioning, operations, performance analysis, trading, grid operation) and for each horizon from seconds to decades. We collect the physical and statistical foundations in one notation and prove four results that govern forecast design: the power curve amplifies relative wind errors by a factor of at least three at partial load; site errors average out only down to a floor set by their mean correlation; averaging two models helps if and only if $k^2 + 2\rho k < 3$ for their RMSE ratio k and error correlation ρ; and under asymmetric imbalance costs the optimal bid is a quantile, not the mean. We review weather-driven price formation, operational and machine-learning weather models, simulations, free data and their licences, and the data-handling errors that most often break forecasts. A case study verifies 92 days of 00 UTC forecasts from ECMWF IFS, ECMWF's machine-learning model AIFS and NOAA GFS against 62 German stations (May to July 2025). At day-ahead lead times AIFS matches IFS for 10 m wind (RMSE 1.36 vs. 1.37 m/s) and beats it for temperature, while GFS trails both. A multi-model mean with a running 30-day bias correction lowers the wind RMSE to 1.12 m/s and extends positive skill against climatology from 5.3 to 8.5 days. All four propositions hold on the data: the median partial-load error elasticity is 3.18, averaging 62 stations removes 73 % of the error as predicted, and bidding the predicted quantile cuts the expected asymmetric cost by 25 % against the point forecast. All data, code and figure tables are published with the paper.

Keywords: renewable energy forecasting; numerical weather prediction; machine-learning weather models; forecast verification; photovoltaics; wind power; electricity markets; probabilistic forecasting

1 Thesis

There is no such thing as *the* weather forecast for renewable energy. A day-ahead trader needs the photovoltaic (PV) output of a whole country in quarter-hours before 12:00 CET. A commissioning engineer needs to know which of the next five days will be clear enough for a capacity test. A crane supervisor needs gusts at hub height for tomorrow morning, an asset manager needs the expected yield of the rest of the year, and a bank needs the production that will be exceeded in nine years out of ten. All five ask the same atmosphere a different question, on a different time scale, with a different cost for being wrong.

The literature treats these questions in separate communities: meteorology studies the skill of numerical weather prediction (NWP) [1], energy forecasting studies the conversion of weather into power and its use in markets [2, 3], and resource assessment studies long-term yields. Practitioners, who have to serve all of them from one data pipeline, are left to assemble the pieces themselves.

The thesis of this paper is that a weather forecast for renewable energy is good only relative to a decision, and that the requirements of every decision can be written down along three axes: the *technology* that converts weather into energy, the *department* that acts on the forecast, and the *horizon* between issue and use. Once those are fixed, the variables, resolution, timing, uncertainty format and scoring rule follow.

The paper makes four contributions.

1. A requirements map: which weather variables matter for each technology (Section 3), what each department across the asset life cycle needs (Section 4) and how horizons, predictability and market deadlines interact (Section 5).

2. The foundations in one notation (Section 6): solar geometry and the PV chain, wind power physics, statistical post-processing, scoring rules, and four propositions that govern forecast design: the amplification of wind errors by the power curve, the smoothing of errors across sites, the condition under which averaging two models helps, and the optimal bid under asymmetric imbalance costs.

3. A practical guide to markets (Section 7), models and simulations (Section 8), free data and their licences, and the data-handling errors that most often break forecasts (Section 9).

4. A case study (Section 10) that tests the propositions on 92

[1] This paper is a research compendium: every figure reads its numbers from a CSV file in `data/processed`, every number quoted from the case study is written by the analysis code in `code/`, and every external fact is listed with its source in `data/literature`. All data used are public. References to legal provisions are for context and are not legal advice.

days of 00 UTC forecasts from ECMWF IFS, ECMWF's machine-learning model AIFS and NOAA GFS, verified against 62 German stations from May to July 2025.

The paper is written as a research compendium. Every figure reads its data from a CSV file, every case-study number in the text is produced by the analysis code, and every external fact is listed with its source. Appendix A explains how to rebuild everything.

2 What a forecast is, and what makes it good

The chain. No operational system forecasts power directly from the sky. Every renewable forecast, whether it comes from a grid operator, a direct marketer, a commercial vendor or a research script, is built from the same chain (Figure 1). Observations from satellites, aircraft, radiosondes and surface stations are combined with a previous forecast by data assimilation into an *analysis*, the best estimate of the current atmosphere. A weather model, either a physics-based NWP model or a machine-learning model trained on reanalysis, steps that state forward. Post-processing removes systematic errors and brings the forecast to the site. A conversion model turns weather into power, using the plant's design and its current state. An uncertainty model attaches quantiles, ensembles or scenarios, and the result feeds a decision: a bid, a schedule, an alert, a report.

Where the error comes from. The dominant error source changes with the horizon. In the first hours, local cloud and wind fluctuations and the plant's state dominate; at day-ahead, the weather model's errors in cloud cover, cloud type and the timing of fronts; beyond about five days, the chaotic growth of initial errors [4], which only ensembles can describe. In practice, the largest *avoidable* errors are often not meteorological: a wrong installed capacity, an inverter that has been off for a week, a curtailment rule the forecast does not know, or a timestamp shifted by one hour (Section 9).

Goodness. Murphy's classic analysis separates three kinds of goodness [5]. *Consistency*: the forecast states what the forecaster actually believes, which for a model means it is optimised for the loss the user faces. *Quality*: the agreement between forecasts and observations, measured by bias, mean absolute error (MAE), root mean square error (RMSE) and, for probabilistic forecasts, proper scoring rules such as the continuous ranked probability score (CRPS) [6]. *Value*: the improvement of the decisions taken with the forecast. A forecast with slightly worse RMSE but better timing of ramps can be worth more to a trader. For energy use, quality and value further require that a forecast is

- specific to the target variable (good 2 m temperature is irrelevant if irradiance under low stratus is poor);
- unbiased across seasons, hours and weather regimes;
- probabilistic and calibrated, then as sharp as possible;
- available before the decision deadline and refreshed when new information arrives;
- at the market's resolution, now 15 minutes in Europe;
- stable from run to run, robust to a missing input, and aware of the plant's state.

Three ways to turn weather into power. A *physical* approach passes NWP irradiance or wind through a model of the plant: solar geometry, transposition, temperature and inverter models for PV; hub-height extrapolation, density correction, power curves and wakes for wind (Section 6). It works from the first day of operation and is explainable, but inherits every NWP bias. A *statistical* approach learns power directly from NWP features and historical output, which corrects systematic errors automatically but needs a clean history and breaks when the plant changes. The *hybrid* approach used by most operational systems feeds the physical model's output into a statistical correction, combining robustness for new plants with the accuracy of learned corrections. Orthogonal to this choice is whether the output is *deterministic*, one value per time step, or *probabilistic*, a distribution. Probabilistic output is now the standard for trading, storage optimisation and reserve sizing because those decisions depend on risk, not only on the expected value [7, 8].

3 Requirements by technology

Each technology converts a different part of the weather into energy, so each needs a different forecast. Table 1 summarises which variables matter; this section explains why.

Photovoltaics. Output follows the irradiance on the plane of the modules (POA), reduced by module temperature, reflection, soiling, snow and the inverter limit. NWP models provide global horizontal irradiance (GHI) and often its direct and diffuse parts; a decomposition model is needed when only GHI is available, and a transposition model turns horizontal components into POA (Section 6). Cloud cover by level and cloud type decide most of the error: low stratus and fog absorb far more than thin cirrus. Aerosols matter more than often assumed: on 4 April 2014 the day-ahead PV forecast for Germany overestimated output by up to 5.3 GW because Saharan dust was missing from the model [9]. Air temperature and wind speed set the module temperature, which costs roughly 0.3–0.4 % of power per kelvin above 25 °C for crystalline silicon. Snow can set output to zero under a clear sky, and on tracker plants gusts and hail trigger stow positions that override the irradiance forecast. Rooftop PV without telemetry appears to grid operators as reduced load, which makes irradiance an input to load forecasting as well.

Concentrating solar. Uses only the direct beam (DNI), which is far more sensitive to aerosols and thin cirrus than GHI. Thermal storage makes the plant dispatchable, so the forecast feeds a dispatch optimisation.

Onshore wind. Output follows the cube of hub-height wind speed between cut-in and rated speed, is flat up to cut-out,

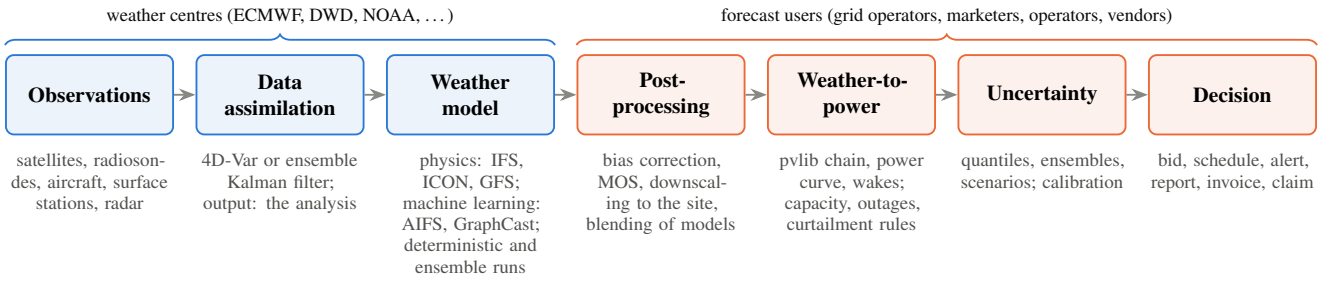


Figure 1: Every renewable forecast is a chain, and every link adds error. Blue stages are run by weather centres and are shared by all users; orange stages are where an energy forecast is made specific to a plant, a portfolio and a decision.

Table 1: Weather variables by technology. Three dots: the variable drives output or operation; two: important correction or constraint; one: minor or indirect; a centred point: not relevant. Storage depends on weather mainly through prices and co-located generation.

Variable	PV	CSP	Onshore wind	Offshore wind	Hydro	Storage	Load and grid
Irradiance (GHI, DNI, DHI)	●●●	●●●	·	·	·	●	●●
Cloud cover by level and type	●●●	●●●	·	·	·	·	●
Aerosols, dust, smoke	●●	●●●	·	·	·	·	·
Air temperature	●●	●	●●	●	●●	●●	●●●
Hub-height wind speed	●	●	●●●	●●●	·	·	●●
Direction, shear, stability, turbulence	·	·	●●	●●●	·	·	·
Gusts and storms	●●	●●	●●	●●	·	·	●●
Air density (pressure, humidity)	·	·	●●	●●	·	·	·
Precipitation	●	·	●	·	●●●	·	●
Snow (fall, depth, melt)	●●	●	●	·	●●●	·	●
Hail and lightning	●●	●●	●	●●	·	·	●●
Icing conditions	·	·	●●	●	·	·	●●
Waves (height, period)	·	·	·	●●●	·	·	·
Fog and visibility	●●	●	●	●●	·	·	·
River flow, soil moisture	·	·	·	·	●●●	·	·

and drops to zero in storms. Hub heights of 100–170 m lie above the 10 m level of most observations, so shear (and its strong day-night cycle over land), veer and atmospheric stability matter. Air density changes output by several percent between summer and winter. Icing near and below 0 °C with fog or freezing rain causes losses and shutdowns. In Germany, permit conditions add weather-dependent curtailments that a trading forecast must reproduce: bat protection on warm, calm nights, shadow flicker on clear days, noise-reduced night modes and bird protection during harvest.

Offshore wind. Adds the marine boundary layer, where low turbulence lets wakes persist for tens of kilometres, sea-surface temperature that controls stability, waves (significant wave height H_s and period) that decide access and installation, and the risk of whole clusters reaching cut-out together.

Hydropower. Weather acts through a catchment and with delay: precipitation, snowpack and its melt, evaporation and upstream flows. Run-of-river plants need river-flow forecasts over hours to two weeks; reservoirs need inflow over weeks to months, which is why Nordic power prices follow the hydrological balance.

Storage and hybrids. Battery storage uses weather indirectly, through price forecasts (Section 7), through the PV or wind output it shares a grid connection with, and through temperature, which drives auxiliary consumption and derating. Hybrid plants need joint forecasts that respect the correlation between technologies and the shared export limit.

Demand and grid. Load depends on temperature (heating and cooling degree days), on cloudiness and on rooftop PV. Overhead-line capacity depends on wind cooling, ambient temperature and solar heating (dynamic line rating), so the windy days that create transport needs also raise capacity.

4 Requirements by department

Across the life of a plant, almost every department uses weather data, but each asks a different question (Table 2). The question fixes the horizon, the data source and what "accurate" means.

Development and financing. Site selection and the energy yield assessment work from the past: 20 or more years of satellite irradiance or reanalysis, corrected to the site with on-site measurements (measure-correlate-predict for wind). The

Table 2: Forecast requirements by department. The horizon is the time between issuing and using the forecast; negative horizons use the past.

Department	Core question	Horizon	Main output	What "good" means
Development	Is the site worth building?	20–40 years of the past	long-term mean, monthly and diurnal profile	unbiased long-term mean
Yield assessment, financing	How much, with what certainty?	1, 10, 20 years	P50, P90, P99	honest uncertainty
Engineering	What must it survive, how should it be sized?	return periods of 50+ years	extremes, clipping, stow settings	correct extremes
Construction	Can we work tomorrow? How many weather days to plan?	0–10 days, plus climatology	go/no-go per activity, weather-downtime statistics	reliable limits, few surprises
Commissioning	When can the tests run, and were they valid?	0–7 days, plus measured data	test windows, weather-corrected results	stable conditions found
Operations, O&M	What do we do today and this week?	0–10 days	alerts, maintenance windows, crew safety	useful warnings, few false alarms
Performance analysis	Is the plant performing?	the past	expected vs. actual, PR, availability, losses	correct expected power
Lost energy, insurance	How much energy and money was or will be lost?	past, and days to weeks ahead	kWh and EUR lost, repair priority	defensible method
Asset management	Will we meet the budget?	month to year	budget, re-forecast, revenue	unbiased P50
Trading, direct marketing	What do we sell, at what risk?	15 min to weeks	production and price quantiles	low imbalance cost
Grid operation	Is the system balanced and secure?	minutes to a week	infeed, load, reserves, congestion	few large misses
PPA and risk	What volume and shape risk do we carry?	years	profiles, capture rates, tails	correct correlations and tails

output is a long-term mean and its uncertainty, expressed as exceedance levels P50, P90 and P99 for one and ten years. Interannual variability is a major component: annual irradiation varies by roughly 3–5 % (one standard deviation) in Central Europe, annual wind energy by more because the power curve amplifies wind-speed variability (Proposition 1). Standards are still consolidating: IEC 61400-15-1 on wind site suitability appeared in 2025 [10], the companion part on yield reporting was in preparation at the time of writing.

Engineering. Design needs extremes rather than averages: 50-year wind speeds and gusts, snow and wind loads, hail return periods, temperature extremes for cable and inverter ratings, and high-resolution irradiance to size the ratio of DC to AC power, because hourly averages hide clipping.

Construction. Forecasts protect schedule, cost and people. Daily decisions compare forecasts with activity limits: hub-height wind and gusts for turbine lifts (light, large loads such as blades are often limited to about 8–10 m/s with conventional cranes, well below crane chart values), wave height for offshore transfers, rain and frost for earthworks and foundations, lightning for electrical work. Months ahead, hindcasts give the probability of weather windows, the consecutive hours below all limits, which feed Monte Carlo schedule simulations and contractual definitions of exceptional weather.

Commissioning. PV capacity tests (IEC TS 61724-2:2025 [11]) need stable, high irradiance; wind power-curve verification (IEC 61400-12-1 [12]) needs enough valid data in every wind-speed bin and direction sector; grid-compliance tests often need the plant above a share of rated power. Short-range forecasts are used to choose test days, and the results are corrected to reference conditions with measured weather.

Operations and maintenance. The control room uses forecasts for severe-weather alerts (storm and hail stow of trackers, icing, lightning, heat derating), crew safety, and to schedule maintenance in low-yield periods: an inverter exchange on an overcast, calm day costs a fraction of the same work on a clear one.

Performance analysis and lost energy. Looking backwards, the expected output under the weather that actually occurred is compared with metered output to compute performance ratios, availability and loss breakdowns. Satellite irradiance is the independent check when on-site sensors drift. The same expected-power model values outages for insurance claims and, in Germany, the curtailed energy (*Ausfallarbeit*) compensated under Redispatch 2.0 [13]. Looking forwards, a forecast turns an open fault into an expected loss until repair, $\sum_t \hat{P}_t \, \Delta t \, \hat{p}_t$, which ranks repairs by money rather than by age.

Asset management. Budgets distribute the long-term P50 over months; year-end re-forecasts combine year-to-date actuals with climatology and, increasingly, sub-seasonal and seasonal outlooks; revenue forecasts need capture prices rather than baseload prices (Section 7).

Trading and direct marketing. The heaviest real-time users:

portfolio production in quarter-hours, refreshed with every model run, with quantiles for bidding and curtailment decisions at negative prices.

Grid operators. Transmission system operators forecast and extrapolate wind and solar infeed for each control area, which the German operators publish [14], and use forecast uncertainty to size reserves and plan redispatch.

PPA structuring and risk. Power purchase agreements transfer volume risk (how much) and shape risk (when). Pricing them needs many weather years combined with price scenarios, and attention to tail events such as long winter periods of low wind and sun.

5 Horizons, predictability and deadlines

Forecasts for renewable energy span nine orders of magnitude in lead time, from seconds for plant controllers to decades for financing (Figure 2). The industry's workhorse is the day-ahead forecast in quarter-hours, followed by updates through the delivery day and outlooks of up to two weeks for planning. Table 3 lists the inputs and methods that dominate each horizon.

How skill decays. Small errors in the initial state grow until the forecast is no better than climatology [4]. For the surface variables that matter for energy, the decay is faster than for the upper-air fields used in headline scores: cloud cover and irradiance lose much of their skill within a few days, 10 m and hub-height wind somewhat later. The case study measures this directly: for daytime 10 m wind over Germany in summer 2025, single models kept positive skill against climatology for 4.3–5.5 days, and a bias-corrected multi-model mean for 8.5 days (Section 10). Beyond the deterministic limit only ensembles carry information, first as probabilities of weather regimes, at sub-seasonal range as weekly anomalies, and at seasonal range as shifted odds, which in Europe are informative mainly in winter.

How accurate is good. Published accuracy depends strongly on normalisation (installed capacity, mean production, daylight hours only) and must be compared with care. Representative German wind farms reached a day-ahead RMSE of 10–15 % of installed capacity, control areas 6–7 % and Germany as a whole 5–6 % in 2007 [15]; nationwide errors were about 3–4 % a decade later [16]. Aggregating 152 PV systems reduced day-ahead MAE by at least 18 % and RMSE by at least 20 % compared with forecasting each system [17]. Both effects, aggregation and steady model improvement, are quantified for the case study in Section 10.

Deadlines. A forecast is only useful if it arrives before the decision. For the European day-ahead auction, which closes at 12:00 CET on the day before delivery, the freshest global runs are those started at 00 UTC, available in the early morning; regional models that run every three hours, and rapid-update runs such as DWD's hourly ICON-D2-RUC [18], add later information if the pipeline is fast. The 12 UTC runs serve the intraday auctions and continuous trading (Figure 4). Since 1 October 2025 the day-ahead market clears in 15-minute units [19], so hourly forecasts no longer match the product.

6 Foundations

This section collects the relations that a forecasting system implements, in one notation, and proves four propositions that govern forecast design. Angles are in degrees, irradiance G in W/m^2, wind speed v in m/s and air density ρ in kg/m^3.

6.1 Solar geometry and the PV chain

With day of year n, latitude φ, declination δ and hour angle ω (zero at solar noon, 15° per hour), the solar zenith angle θ_z, the extraterrestrial irradiance and the air mass are

$$\cos\theta_z = \sin\varphi\sin\delta + \cos\varphi\cos\delta\cos\omega, \tag{1}$$

$$G_0 = G_{sc}\left(1 + 0.033\cos\tfrac{360^\circ n}{365}\right)\cos\theta_z, \tag{2}$$

$$AM = \left[\cos\theta_z + 0.50572\,(96.07995 - \theta_z)^{-1.6364}\right]^{-1}, \tag{3}$$

with $G_{sc} \approx 1361$ W/m^2 and the air mass of Kasten and Young [20]. Operational code uses the NREL solar position algorithm [21]. Global, direct normal and diffuse irradiance are linked by $GHI = DNI\cos\theta_z + DHI$. Two normalisations remove the deterministic cycle: the clearness index $k_t = GHI/G_0$ and the clear-sky index $k_c = GHI/GHI_{\text{clear}}$, where GHI_{clear} comes from a clear-sky model such as Ineichen-Perez with Linke turbidity [22]. Forecasting k_c instead of GHI leaves only the weather to be learned. When a model gives only GHI, a decomposition model estimates the diffuse fraction $k_d = DHI/GHI$; the Erbs model [23] is

$$k_d = \begin{cases} 1 - 0.09k_t, & k_t \le 0.22, \\ 0.9511 - 0.1604k_t + 4.388k_t^2 \\ \quad -16.638k_t^3 + 12.336k_t^4, & 0.22 < k_t \le 0.80, \\ 0.165, & k_t > 0.80. \end{cases} \tag{4}$$

For a surface with tilt β and azimuth γ, the angle of incidence θ and the plane-of-array irradiance are

$$\cos\theta = \cos\theta_z\cos\beta + \sin\theta_z\sin\beta\cos(\gamma_s - \gamma), \tag{5}$$

$$G_{\text{POA}} = DNI\max(0,\cos\theta) + G_d + GHI\,\rho_g\tfrac{1-\cos\beta}{2}, \tag{6}$$

where ρ_g is the ground albedo and the sky-diffuse part G_d is $DHI(1+\cos\beta)/2$ in the isotropic model and is computed with the anisotropic Perez model [24] in practice. Reflection losses follow an incidence-angle modifier, for example IAM $= 1 - b_0(1/\cos\theta - 1)$ with $b_0 \approx 0.05$. The cell temperature (Faiman [25]), DC power (PVWatts [26]) and AC power with

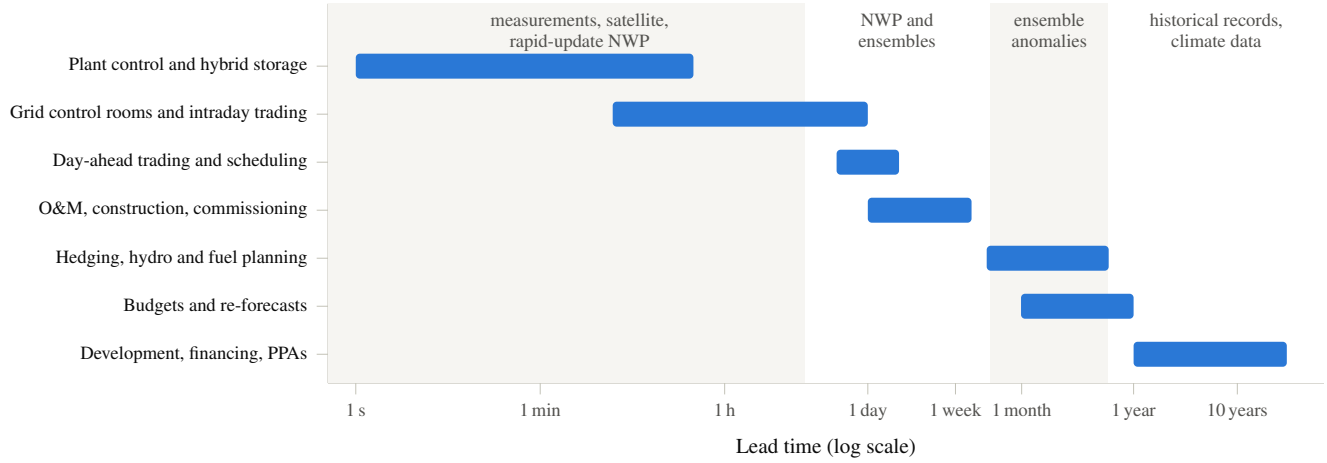


Figure 2: Who forecasts how far ahead. Each bar spans the lead times a user group typically works with; shaded bands show which information dominates. Performance analysis, lost-energy calculations and insurance claims use the past and are not shown.

clipping are

$$T_c = T_a + \frac{G_{\mathrm{POA}}}{U_0 + U_1 v}, \tag{7}$$

$$P_{DC} = P_{DC,0}\,\frac{G_{\mathrm{POA,eff}}}{1000}\,\big[1 + \gamma(T_c - 25)\big], \tag{8}$$

$$P_{AC} = \min\big(\eta\, P_{DC},\; P_{AC,\max}\big), \tag{9}$$

with $U_0 = 25\,\mathrm{W/(m^2K)}$, $U_1 = 6.84\,\mathrm{W\,s/(m^3K)}$ and $\gamma \approx -0.0035\,\mathrm{K^{-1}}$ for current crystalline modules. All steps are implemented in pvlib [27].

Figure 3 runs this chain for a 10 MWp plant near Leipzig with a DC/AC ratio of 1.3 on a clear 21 June 2025, using observed air temperature and wind. Tilting the modules by 20° raises the daily irradiation from 7.66 to 8.03 kWh/m^2; module temperatures of up to 56 °C at an air temperature of 29 °C cost 6.8 % of the DC energy, and the inverter limit clips the output for 3.7 hours around solar noon (11:10 UTC), losing 2.8 % of the AC energy. On such a day the inverter limit, not the irradiance forecast, decides the midday output.

6.2 Wind power

The power extracted by a rotor of swept area A is

$$P = \tfrac{1}{2}\rho A C_p v^3, \qquad C_p \le \tfrac{16}{27} \approx 0.593, \tag{10}$$

the upper bound being the Betz limit [28]. Air density follows from pressure and temperature, $\rho = p/(R_d T)$ with $R_d = 287.05\,\mathrm{J/(kg\,K)}$, and power curves measured at a reference density ρ_0 are applied to the density-normalised speed $v_n = v(\rho/\rho_0)^{1/3}$ [12]. Winds at 10 m or 100 m are brought to hub height z with a power law $v(z) = v(z_r)(z/z_r)^{\alpha}$, where $\alpha \approx 0.14$ in neutral conditions and much more on stable nights, or with the logarithmic profile $v(z) = (u_*/\kappa)[\ln(z/z_0) - \psi_m(z/L)]$ including a stability correction. The long-term energy is $\mathrm{AEP} = 8760 \int P(v) f(v)\,dv$ with a Weibull density $f(v) = \frac{k}{c}(\frac{v}{c})^{k-1} e^{-(v/c)^k}$, and wake losses can be estimated with the Jensen model [29], $v_w/v_\infty = 1 - (1 - \sqrt{1 - C_T})/(1 + 2k_w x/D)^2$.

Proposition 1 (Power-curve amplification)**.** *Let $P(v) = k\,(v^3 - v_{ci}^3)$ between the cut-in speed v_{ci} and the rated speed v_r, and $P = P_r$ between v_r and cut-out. The elasticity of power with respect to wind speed is*

$$\eta(v) = \frac{v}{P}\frac{dP}{dv} = \frac{3v^3}{v^3 - v_{ci}^3} \ge 3 \quad \text{on } (v_{ci}, v_r), \tag{11}$$

and $\eta = 0$ between rated and cut-out speed. A small relative wind-speed error ε therefore becomes a relative power error of at least 3ε at partial load and of none at full load.

Proof. On (v_{ci}, v_r), $dP/dv = 3kv^2$, so $\eta = 3kv^3/(k(v^3 - v_{ci}^3))$, which exceeds 3 because $v_{ci}^3 > 0$. On (v_r, v_{co}) the derivative is zero. □

Wind forecasts are therefore hardest to use exactly where turbines spend most of their time, at partial load, and errors near cut-in are amplified most. The case study measures an aggregate amplification of 1.9 (Section 10).

6.3 Hydro, demand and grid

Hydropower is $P = \eta \rho_w g Q H$ for flow Q and head H; inflow follows a catchment water balance $dS/dt = P_{\mathrm{precip}} - ET - Q$, with snowmelt often modelled by a degree-day factor, $M = \mathrm{DDF}\max(0, T - T_0)$. Temperature-driven demand is summarised by heating and cooling degree days, $\mathrm{HDD} = \sum \max(0, T_b - \bar{T})$ and $\mathrm{CDD} = \sum \max(0, \bar{T} - T_b)$. Dynamic line rating solves the conductor heat balance $q_c(v, T_a) + q_r = q_s + I^2 R(T_c)$ for the current I that keeps the conductor at its limit temperature.

6.4 Statistical post-processing and uncertainty

Model output statistics (MOS) [30] regress the target on NWP outputs, $\hat{y} = a + \sum_i b_i x_i$, per site and lead time; recursive variants such as a Kalman-filtered bias $\hat{b}_t = \hat{b}_{t-1} + K_t(e_t - \hat{b}_{t-1})$

Table 3: Forecast horizons, their inputs and their users.

Horizon	Lead time	Resolution	Main inputs	Typical methods	Main users
Very short term	seconds to 30 min	1 s–1 min	sky cameras, live plant data, nacelle or lidar wind	persistence, image-based cloud tracking	plant controllers, hybrid PV and storage
Nowcasting	30 min to 6 h	5–15 min	satellite images, live SCADA, rapid-update NWP, radar	cloud motion vectors, machine learning on images, blending	intraday traders, grid control rooms
Intraday	1 to 24 h	15 min	latest NWP runs, nowcasts, live data	statistical blending, frequent re-runs	intraday auctions and continuous trading
Day-ahead	12 to 48 h	15 min	00 and 06 UTC runs of several models	physical model plus statistical correction, quantiles	day-ahead auction, scheduling, O&M planning
Medium range	2 to 15 days	1 h–1 day	global NWP and ensembles	calibrated ensemble statistics	maintenance, construction, weekly hedging
Sub-seasonal	2 to 6 weeks	weekly	extended-range ensembles	weekly anomalies, regime probabilities	monthly forwards, hydro, gas storage
Seasonal	1 to 7 months	monthly	multi-system seasonal forecasts	tercile probabilities	budgets, reservoirs, winter adequacy
Long term	1 to 30+ years	hourly to annual	reanalysis, satellite records, on-site data, climate projections	long-term correction, P-values, Monte Carlo	development, financing, PPAs

follow slow drifts. Probabilistic forecasts take three common forms. Ensemble model output statistics (EMOS) [31] fit a distribution to the ensemble mean $\bar{x}$ and variance s^2,

$$Y \mid X \sim \mathcal{N}\left(a + b\bar{x},\; c + ds^2\right), \tag{12}$$

by minimising the CRPS (with truncated or censored variants for bounded power). Quantile regression [32] learns each quantile q_τ directly by minimising the pinball loss, with linear models, gradient boosting or neural networks. Conformal prediction wraps any point model with intervals $\hat{y} \pm q_{1-\alpha}(|r|)$ calibrated on held-out residuals r. Post-processing is reviewed in [33].

6.5 Scoring

For errors $e_t = \hat{y}_t - y_t$, bias, MAE and RMSE are $\bar{e}$, $\overline{|e|}$ and $\sqrt{\overline{e^2}}$, with $\mathrm{RMSE}^2 = \bar{e}^2 + \sigma_e^2$ splitting systematic from random error. The MSE skill score against a reference is $\mathrm{SS} = 1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{ref}}$, positive when the forecast beats the reference. For a quantile q at level τ and for a predictive distribution F,

$$L_\tau(y, q) = \begin{cases} \tau(y - q), & y \geq q, \\ (1 - \tau)(q - y), & y < q, \end{cases} \tag{13}$$

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} \left(F(x) - \mathbf{1}\{x \geq y\}\right)^2 dx, \tag{14}$$

both proper [6]; for an ensemble $x_1, \ldots, x_M$, $\mathrm{CRPS} = \frac{1}{M}\sum_i |x_i - y| - \frac{1}{2M^2}\sum_{i,j} |x_i - x_j|$ [34]. MAE is minimised by the median, RMSE by the mean and the pinball loss by the τ-quantile, so the scoring rule must match the decision (Proposition 4). Differences between two forecasts are tested with the Diebold-Mariano test [35]. Percentage errors such as MAPE are unusable for PV and wind because the target is often zero.

6.6 Four design results

Proposition 2 (Geographic smoothing)**.** *Let $e_1, \ldots, e_N$ be site errors with mean zero, common variance σ^2 and mean pairwise correlation $\bar{r}$. The error of the site average has standard deviation*

$$\sigma_{\bar{e}} = \sigma\sqrt{\frac{1 + (N-1)\,\bar{r}}{N}} \xrightarrow[N \to \infty]{} \sigma\sqrt{\bar{r}}. \tag{15}$$

Proof. $\mathrm{Var}\left(\sum_i e_i\right) = \sum_i \sigma^2 + \sum_{i \neq j} \bar{r}\sigma^2 = N\sigma^2 + N(N-1)\bar{r}\sigma^2$; divide by N^2. □

Portfolio and national forecasts are therefore much more accurate than single sites, but never better than $\sigma\sqrt{\bar{r}}$: the part of the error shared by all sites, typically the timing of a front, cannot be diversified away.

Proposition 3 (When averaging two models helps)**.** *Let two forecasts have errors with root mean squares $\sigma_1 \leq \sigma_2$ and correlation $\rho = \overline{e_1 e_2}/(\sigma_1\sigma_2)$, and let $k = \sigma_2/\sigma_1$. The equal-weight average has a lower mean squared error than the better forecast if and only if*

$$k^2 + 2\rho k < 3. \tag{16}$$

Proof. The average has error $(e_1 + e_2)/2$ and mean squared error $(\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2)/4$, which is below σ_1^2 exactly when $\sigma_2^2 + 2\rho\sigma_1\sigma_2 < 3\sigma_1^2$; divide by σ_1^2. □

For highly correlated errors ($\rho = 0.8$) the average helps only if the weaker model is within about 11 % of the better one ($k < 1.11$); for $\rho = 0.5$ the margin grows to 30 %. Blending pays when models are similarly good and fail differently, the classic result of forecast combination [36].

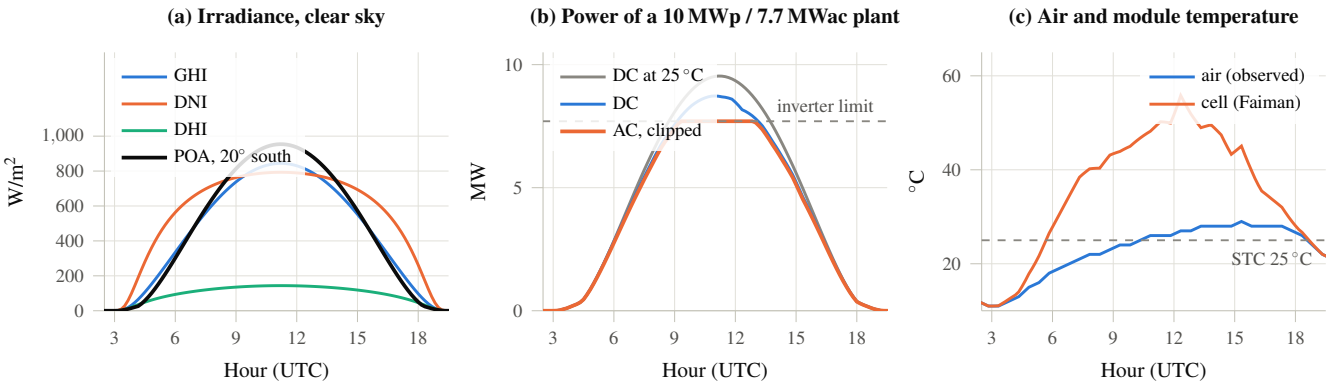


Figure 3: On a clear day, temperature and the inverter limit shape PV output as much as irradiance. A 10 MWp fixed-tilt plant near Leipzig on 21 June 2025 (clear-sky irradiance from the Ineichen-Perez model, observed air temperature and wind at Leipzig-Halle airport). (a) Transposition to the 20° south-facing plane raises the peak above GHI. (b) Module heating lowers DC power below its 25 °C value, and the 7.7 MW inverter limit clips the midday output. (c) At noon the modules run more than 20 K above air temperature.

Proposition 4 (Optimal bid under asymmetric costs)**.** *Let production Y have distribution function F, and let each unit of shortfall below the bid b cost c_s and each unit of surplus cost c_l. The expected cost $\mathbb{E}[c_s(b-Y)^+ + c_l(Y-b)^+]$ is minimised by*

$$b^* = F^{-1}\left(\frac{c_l}{c_l + c_s}\right). \tag{17}$$

Proof. The derivative of the expected cost with respect to b is $c_s F(b) - c_l(1-F(b))$, which is increasing in b and zero at $F(b) = c_l/(c_l+c_s)$. □

This is the newsvendor result applied to energy trading [37]: if a shortfall costs three times as much as a surplus, the expected-cost-minimising bid is the 25 % quantile, not the median. A point forecast cannot express that; a probabilistic forecast can.

Lemma 1 (Time-averaged model fields)**.** *If a model reports a flux as its mean $\bar{A}_t$ over $[0,t]$ since the start of the run, the mean over $[t-\Delta,t]$ is $(t\bar{A}_t - (t-\Delta)\bar{A}_{t-\Delta})/\Delta$. If it reports an accumulation S_t in J/m², the mean flux is $(S_t - S_{t-\Delta})/\Delta$ with Δ in seconds.*

Proof. $t\bar{A}_t$ and S_t are the integrals of the flux over $[0,t]$; subtracting the integral over $[0,t-\Delta]$ leaves the integral over the interval, which is divided by its length. □

Section 9 lists which operational models use which convention.

6.7 Yield and performance

Exceedance levels follow, under a normal approximation, from the P50 and the total relative uncertainty, $P_X = P_{50}(1 - z_X\sigma_{\text{tot}})$ with $z_{90} = 1.282$ and $z_{99} = 2.326$; independent components add in quadrature, and the interannual component of an N-year mean shrinks as $\sigma_{\text{IAV}}/\sqrt{N}$. The performance ratio is $\text{PR} = (E_{AC}/P_{STC})/(H_{\text{POA}}/G_{STC})$ with $G_{STC} = 1000\,\text{W/m}^2$ [38]. Lost energy during an outage is $E_{\text{lost}} = \sum_t \max(0, P_t^{\text{exp}} - P_t^{\text{meas}})\,\Delta t$ with the expected power taken from a reference, and energy-based availability is $E_{\text{actual}}/(E_{\text{actual}} + E_{\text{lost}})$.

7 Weather and electricity prices

In power systems with a large share of wind and solar, the weather forecast is the main driver of short-term prices, and traders react to every new model run.

Merit order and residual load. Power plants are dispatched in order of marginal cost, and the price is set by the most expensive unit still needed. Wind and solar have almost no marginal cost, so they enter first and push gas and coal units out of the market. Prices therefore follow the residual load,

$$L_t^{\text{res}} = L_t - W_t - S_t, \tag{18}$$

the load L that remains after wind W and solar S, and weather acts on both sides: wind and irradiance on supply, temperature and daylight on demand. Strong wind lowers prices around the clock; sunshine creates a midday trough and an evening ramp; sun and wind on spring holidays produce negative prices; long, dark, calm winter spells (*Dunkelflaute*) let gas set the price. Hydro-dominated markets follow precipitation and snowpack over months.

Evidence. Table 4 lists recent German observations. Negative day-ahead prices occurred in 573 hours of 2025 [39, 40]; the solar capture rate, the solar-weighted price divided by the baseload price, fell to 29.7 % in May 2025 [41]; and during the Dunkelflaute of 12 December 2024 the hourly day-ahead price reached 936 EUR/MWh, a peak that the regulator found no evidence of abuse in [42].

The market timeline. Figure 4 shows the European sequence for one delivery day. Futures cover years to days ahead; the single day-ahead coupling (SDAC) auction closes at 12:00 CET on the day before delivery and has cleared 15-minute products since 1 October 2025 [19]; three pan-European intraday auctions (15:00 and 22:00 on D-1, 10:00

Table 4: Weather leaves visible marks on German power prices. Source data with links: `data/literature/market_facts.csv`.

Observation	Value	Source
Hours with negative day-ahead prices, 2025	573 h	[39]
Solar capture rate, May 2025	29.7 %	[41]
Share of potential solar output at negative prices, Jan–May 2025	28 %	[43]
Highest hourly day-ahead price, Dunkelflaute of 12 Dec. 2024	936 EUR/MWh	[42]
Day-ahead PV overestimate in a Saharan dust outbreak, 4 Apr. 2014	5.3 GW	[9]

on D) have run since 13 June 2024 [44]; continuous intraday trading runs until shortly before delivery; and remaining deviations are settled at the imbalance price every 15 minutes. Each gate is a decision point that a forecast pipeline has to serve with the freshest model run available at that moment.

Imbalance. A seller whose plant produces q_t^{act} after selling q_t^{sold} settles the difference at the imbalance price, at a cost relative to the day-ahead price of approximately

$$C^{\text{imb}} = \sum_t \left(q_t^{\text{sold}} - q_t^{\text{act}}\right)\left(p_t^{\text{imb}} - p_t^{\text{DA}}\right). \tag{19}$$

Two features make this costlier than a naive estimate. Forecast errors are correlated across the market (Proposition 2: the shared part does not average out), so when wind across the country falls short, nearly every wind seller is short at once and the imbalance price rises against them. And because short and long positions are priced differently, the expected-cost-minimising bid is a quantile, not the expected value (Proposition 4).

Germany. Most wind and solar plants sell through a direct marketer (*Direktvermarkter*) who forecasts, trades and carries the imbalance risk, while the plant receives a sliding market premium relative to a monthly, technology-specific market value. For PV commissioned since 25 February 2025, no premium is paid in negative-price quarter-hours [45], so a production forecast without a price forecast no longer predicts revenue. Grid-related curtailment (Redispatch 2.0) is compensated on the basis of calculated lost energy [13].

Price forecasting. Fundamental models build the supply stack from plant availability, fuel and carbon prices and intersect it with the residual load. Statistical and machine-learning models regress prices on forecasts of load, wind and solar for the zone and its neighbours, fuel and carbon prices, calendar effects and lagged prices; the open benchmark of Lago et al. [46] and the review by Weron [47] are good starting points. Weather-driven price information is strongest for the next one or two days, remains useful for about a week through ensembles, and beyond that blends into climatology and fundamentals.

8 Models, methods and simulations

Physics-based NWP. An NWP model solves the equations of motion, mass, energy and moisture on a global or regional grid, starting from an analysis built by data assimilation (4D-Var at ECMWF, ensemble Kalman filters in DWD's ICON system). Processes smaller than a grid box, such as radiation, cloud microphysics, convection and boundary-layer turbulence, are parameterised, and for energy these schemes are where most errors originate. Ensembles rerun the model from perturbed states to describe uncertainty. Regional models at 1–3 km resolve convection explicitly. Table 5 lists the main systems available as open data.

Machine-learning weather models. Since 2023, models trained on decades of ERA5 reanalysis [48] match or beat physics models on many headline scores while running in minutes on a single accelerator: GraphCast [49], GenCast [50], Aurora [51] and ECMWF's AIFS [52], whose deterministic and ensemble versions became operational in 2025 [53]. NOAA followed in December 2025 with AIGFS, AIGEFS and a 62-member hybrid physics-AI ensemble [54], and Google's WeatherNext 2 generates large ensembles with hourly output [55]. For energy users three caveats matter. Many models output only a subset of variables, so surface radiation, cloud cover and 100 m wind must be checked; most run at about 0.25° with 6-hourly steps; and deterministic versions smooth extremes at longer lead times. WeatherBench 2 provides common verification [56]. The case study (Section 10) compares AIFS with IFS and GFS at the station level.

Nowcasting. For the first hours, observations beat models. Geostationary satellites (Meteosat Third Generation has scanned Europe operationally since December 2024) give cloud fields every few minutes, which are converted to irradiance and advected with cloud motion vectors or learned models; sky cameras cover the next minutes at a single plant; radar tracks precipitation and convective cells; and rapid-update NWP such as ICON-D2-RUC refreshes hourly. Operational systems blend these sources with live plant data, shifting weight from measurements to NWP as the lead time grows.

From weather to power. The physical chain of Section 6 is implemented in pvlib [27] for PV and in windpowerlib [57], PyWake or FLORIS for wind. The statistical layer ranges from linear MOS through gradient-boosted trees, the workhorse for tabular NWP features, to neural networks on NWP fields and graph networks across many sites. Typical inputs are several NWP sources, clear-sky irradiance and solar angles, lead time, calendar terms, ensemble statistics and, for intraday use, recent measurements; targets are normalised by capacity or clear-sky power. Portfolio and regional forecasts upscale from reference plants and reconcile plant, portfolio and regional forecasts so that they add up.

Probabilistic forecasts. Raw ensembles passed through a power model are physically consistent but usually underdispersive, so they are calibrated with EMOS or Bayesian model averaging [31]. Quantile regression and quantile forests

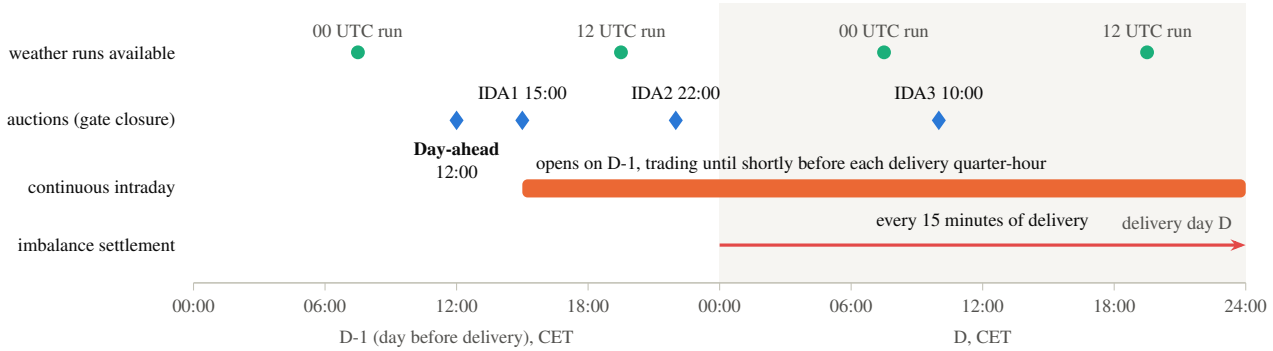


Figure 4: A delivery day has many deadlines, and each should use the freshest forecast. European short-term markets for delivery day D, in CET. Diamonds mark auction gate closures; the orange bar is continuous intraday trading; green dots show roughly when the global 00 and 12 UTC model runs become available (a few hours after their start, depending on the provider). Since 1 October 2025 the day-ahead auction trades 15-minute products.

Table 5: Operational weather models available as open data (status November 2026). Grid spacing is approximate; open-data products are sometimes coarser than the native model. Source data: `data/literature/nwp_models.csv`.

Model	Centre	Type	Grid	Range	Runs/day	Licence
IFS HRES and ENS	ECMWF	physics; ENS 51 members	9 km (open 0.25°)	10 / 15 days	4	CC-BY-4.0
AIFS Single and ENS	ECMWF	machine learning	about 0.25°	15 days	4	CC-BY-4.0
Extended range	ECMWF	physics ensemble	ensemble	46 days	1	CC-BY-4.0 (partly open)
GFS and GEFS	NOAA	physics; GEFS 31 members	0.25°	16 days	4	public domain
AIGFS, AIGEFS, HGEFS	NOAA	machine learning and hybrid	about 0.25°	16 days	4	public domain
ICON, ICON-EU	DWD	physics, with ensembles	13 km, 6.5 km	7.5, 5 days	4, 8	open, with attribution
ICON-D2 (EPS, RUC)	DWD	convection-permitting	about 2 km	2 days (RUC 14 h)	8 (RUC 24)	open, with attribution
MOSMIX	DWD	statistical post-processing	5,400 stations	10 days	several	open, with attribution
AROME, ARPEGE	Météo-France	physics	1.3 km, 5 km	2, 4 days	several	Etalab open licence
HRRR	NOAA	rapid refresh	3 km (USA)	18–48 h	24	public domain
CAMS global	Copernicus	aerosols and dust	about 40 km	5 days	2	Copernicus licence

learn quantiles directly; distributional models predict full densities; analog ensembles reuse what happened after similar past forecasts; conformal prediction adds calibrated intervals to any model. Storage optimisation and stochastic dispatch need scenarios that respect temporal correlation, typically generated with a Gaussian copula over marginal quantiles.

Combining models. Because NWP systems fail differently, blending several sources usually beats the best single source, with static, lead-dependent or performance-based weights. Proposition 3 states when even an equal-weight average of two models helps.

Simulations. Beyond forecasting, the industry relies on a family of simulations that use the same weather data: energy-yield simulation (PVsyst, SAM, pvlib, WAsP, WindPRO); wake and flow simulation from engineering models to computational fluid dynamics and large-eddy simulation; mesoscale downscaling with WRF or ICON for wind atlases; Monte Carlo propagation of uncertainty to P90 and revenue at risk; resampling of 30 or more historical weather years for budgets, PPAs and adequacy studies; weather-window simulation for construction and offshore logistics; hydrological rainfall-runoff and snowmelt models; icing, snow and soiling loss models; power-system and market models such as PyPSA or Antares for price scenarios; and linear or mixed-integer optimisation of storage dispatch under forecast scenarios.

9 Data: sources, licences and pitfalls

Free sources. Almost everything needed to build a forecasting system is available free of charge (Table 6). Real-time forecasts come from ECMWF, whose whole real-time catalogue has been open under CC-BY-4.0 since 1 October 2025 [58],

from NOAA, DWD, Météo-France and MET Norway. Reanalyses (ERA5 [48], CERRA) and satellite irradiance records (SARAH-3, CAMS, NSRDB, PVGIS) provide the long history; national weather services and NOAA's Integrated Surface Database [59] provide observations; the ENTSO-E Transparency Platform, SMARD, Energy-Charts and the German transmission system operators provide generation, forecasts and prices that serve both as targets and as benchmarks.

Licences. Commercial use is the dividing line. NOAA data are public domain; ECMWF, DWD and Copernicus data can be used commercially with attribution; the free tier of the popular Open-Meteo API is restricted to non-commercial use [60], as are TIGGE and Renewables.ninja. A system built for a company should either pay for a commercial tier or run directly on the open primary sources.

Training data must be as issued. A model that will run on forecasts must be trained on forecasts, with the lead time and availability it will have in operation. Training on reanalysis or observations as inputs teaches a relationship that does not exist operationally, because real inputs are worse. Archives of past forecasts exist (NOAA's GFS, HRRR and GEFS reforecast archives, ECMWF open-data mirrors, Open-Meteo's historical-forecast and previous-run interfaces), but for many regional models the only archive is the one a user builds: DWD keeps ICON files on its open-data server for about a day.

Time conventions. Store everything in UTC and record, for every source, whether a time stamp marks the start, the end or the middle of an interval, or an instant. NWP radiation is rarely instantaneous: ECMWF reports accumulations in J/m^2 since the start of the run, DWD's ICON reports means since the start of the run, GFS alternates 3- and 6-hour averaging windows, and Open-Meteo returns means over the preceding hour. Lemma 1 converts the first two into interval means; getting it wrong shifts or scales the forecast systematically, most visibly in the morning and evening ramps. Solar position for an interval mean is computed at the interval's midpoint.

Plant state and sensors. Training data must be filtered for curtailment and outages, or the model learns to under-forecast; installed capacity must be tracked through extensions and repowering; clipping and export limits should be modelled explicitly. Pyranometers drift, soil and misalign; reference cells and pyranometers respond differently to spectrum and angle; nacelle anemometers sit behind the rotor. Independent satellite irradiance and clear-sky curves are the cheapest quality checks.

Evaluation. Backtests must roll forward in time (never randomly shuffle time series), cover at least one full year where possible, report errors by lead time and weather regime, compare against persistence and climatology, state the normalisation, and test differences for significance. Monitoring continues in operation: model upgrades at the weather centres shift biases overnight.

10 Case study: three global models over Germany

The case study measures how quickly skill decays, what simple post-processing adds, and whether the four propositions of Section 6 hold on real forecasts. It uses only public data, and every number below is written by the analysis code (Appendix A).

10.1 Data and design

Forecasts. The 00 UTC runs of 1 May to 31 July 2025 (92 runs per model) of ECMWF IFS and ECMWF AIFS Single from ECMWF open data [58], and of NOAA GFS from the NOAA archive [61], all on 0.25° grids, at lead times of 0 to 240 h in 12-hour steps. For each run and lead time only the 10 m wind components and 2 m temperature were downloaded, with byte-range requests on the GRIB2 indexes, and interpolated bilinearly to the stations; every download is logged with its SHA-256 hash.

Observations. Hourly reports of the NOAA Integrated Surface Database [59] from German stations below 600 m. The report closest to the valid time within 30 minutes is used; stations are kept if both variables are available at 90 % or more of the valid times, which leaves 62 stations, mostly airports, spread over the whole country.

Forecasts compared. The three raw models; their multi-model mean (MMM), the plain average; the same four after a running bias correction (BC) that subtracts, for each model, station and lead time, the mean error of the previous 30 runs whose observations were already available when the run started; and two references: persistence, the latest observation at the same time of day available at the start of the run, and the 2020–2024 station climatology for the same month and hour, which is independent of the test data.

Verification. Scores are computed for the 61 runs from 1 June (May only spins up the bias correction), giving 231,048 model-observation pairs per variable. Intervals are 95 % bootstrap intervals that resample whole runs, which keeps the correlation between stations [62]; differences are tested with the Diebold-Mariano test on daily mean squared errors [35]. Valid times alternate between 12 UTC (half-day lead times) and 00 UTC (whole days); the figures show the 12 UTC valid times, when solar output and wind ramps matter most, and the data files contain both.

10.2 Skill decays within days, and post-processing buys days

Figure 5 and Table 7 summarise the results. At day-ahead lead times the raw IFS and AIFS forecasts of 10 m wind are statistically indistinguishable (RMSE 1.37 and 1.36 m/s, Diebold-Mariano $p = 0.45$), GFS is worse (1.46 m/s, $p < 0.001$), and all three beat persistence (2.16 m/s) and climatology (1.66 m/s). For 2 m temperature AIFS is slightly better than IFS (1.51 vs. 1.56 K, $p = 0.03$) and GFS clearly worse.

Table 6: Free data for weather-based energy forecasting. The last column states whether commercial use is permitted. Full list with URLs: `data/literature/data_sources.csv`.

Purpose	Source	Content	Commercial use
Real-time forecasts	ECMWF open data	IFS and AIFS, deterministic and ensemble, 0.25°	yes (CC-BY-4.0)
	NOAA GFS, GEFS, AIGFS	global forecasts and ensembles	yes (public domain)
	DWD ICON, ICON-D2, MOSMIX	fields down to 2 km; station point forecasts	yes (attribution)
	Météo-France, MET Norway	AROME, ARPEGE, MEPS, point API	yes (open licences)
	CAMS	aerosol and dust forecasts	yes (Copernicus licence)
Convenience API	Open-Meteo	forecasts, past forecasts, satellite radiation, ERA5	free tier: no
Past forecasts	NOAA archives on AWS and Google Cloud; TIGGE	GFS, HRRR, GEFS reforecasts; multi-centre ensembles	NOAA: yes; TIGGE: no
Reanalysis	ERA5, ERA5-Land, CERRA, COSMO-REA6	hourly gap-free weather since 1940 (ERA5)	yes
Satellite irradiance	SARAH-3, CAMS radiation, PVGIS, NSRDB	irradiance records and PV simulation	yes (attribution)
Observations	DWD Climate Data Center, NOAA ISD, BSRN	station weather and reference radiation	DWD, ISD: yes; BSRN: research
Energy and markets	ENTSO-E, SMARD, Energy-Charts, netztransparenz.de	generation, TSO forecasts, prices, imbalance	generally yes
Plant data	Kelmarsh and Penmanshiel (Zenodo)	10-minute SCADA of two UK wind farms	yes (CC-BY-4.0)
Resource atlases	Global Solar Atlas, Global Wind Atlas; Renewables.ninja	long-term maps; simulated time series	atlases: yes; ninja: no

The raw wind forecasts carry a large systematic error with a day-night cycle: at 12 UTC on the second day the models underestimate the observed wind by -0.57 (IFS) and -0.77 m/s (AIFS), at 00 UTC they overestimate it by about 0.34 m/s, a typical signature of near-surface stability and of the difference between a 0.25° grid box and an airport anemometer. The AIFS underestimate grows with lead time, to -1.23 m/s at 9.5 days, consistent with the smoothing of deterministic machine-learning forecasts. The 30-day bias correction removes most of this error: it lowers the day-ahead wind RMSE of the multi-model mean by 14 %, from 1.30 to 1.12 m/s ($p < 0.001$), and the temperature RMSE by 8 %.

Skill against climatology (Figure 5b) defines the useful horizon. For daytime 10 m wind it reaches zero after 4.3 days for GFS, 5.3 for IFS and 5.5 for AIFS; averaging extends it to 6.3 days and bias correction of the average to 8.5 days. Temperature keeps positive skill for longer, beyond 9.5 days for AIFS and the multi-model mean. Model averaging and a simple bias correction together thus extend the useful daytime wind forecast by about three days, without any change to the weather models.

Model averaging (Proposition 3). The day-ahead wind errors of IFS and AIFS have a correlation of 0.90 and an RMSE ratio of 1.02, so $k^2 + 2\rho k$ lies just below 3 and averaging helps, but only a little: the three-model mean is 4 % better than the best single model. The error correlation falls with lead time, from 0.92 at 12 h to 0.68 at 10 days, so the benefit of averaging grows with lead time, as Figure 5 shows. After bias correction AIFS pulls ahead of IFS (RMSE ratio up to 1.17), and at 4 of 21 wind lead times the condition fails: there the IFS-AIFS average is worse than AIFS alone, exactly as the proposition predicts.

10.3 Errors that average out, and errors that do not

Proposition 2 predicts how much a portfolio gains from geographic spread. The day-ahead wind errors of the bias-corrected multi-model mean are correlated at 0.18 on average for stations less than 100 km apart and at 0.02 beyond 500 km, with a mean of 0.06 over all 1,891 pairs (Figure 6a). Averaging the errors of all 62 stations leaves 0.27 of the single-station RMSE, a reduction of 73 %, and the equal-variance formula (15) matches the empirical curve at every portfolio size (Figure 6b). With so low a mean correlation the floor $\sqrt{\bar{r}} = 0.24$ is reached only slowly. Errors of 10 m wind at single stations are dominated by local effects; hub-height wind and irradiance errors over a region are more strongly correlated, so real portfolios reach their floor sooner.

10.4 From wind error to power error and to the bid

To isolate the effect of the power curve, observed and forecast 10 m winds were scaled to a pseudo hub height of 120 m with a neutral power law and passed through the power curve of a V126/3450 turbine (3,450 kW) from the OpenEnergy turbine library [57]. The resulting pseudo power is not the output of any real farm, but forecast and observation are transformed identically.

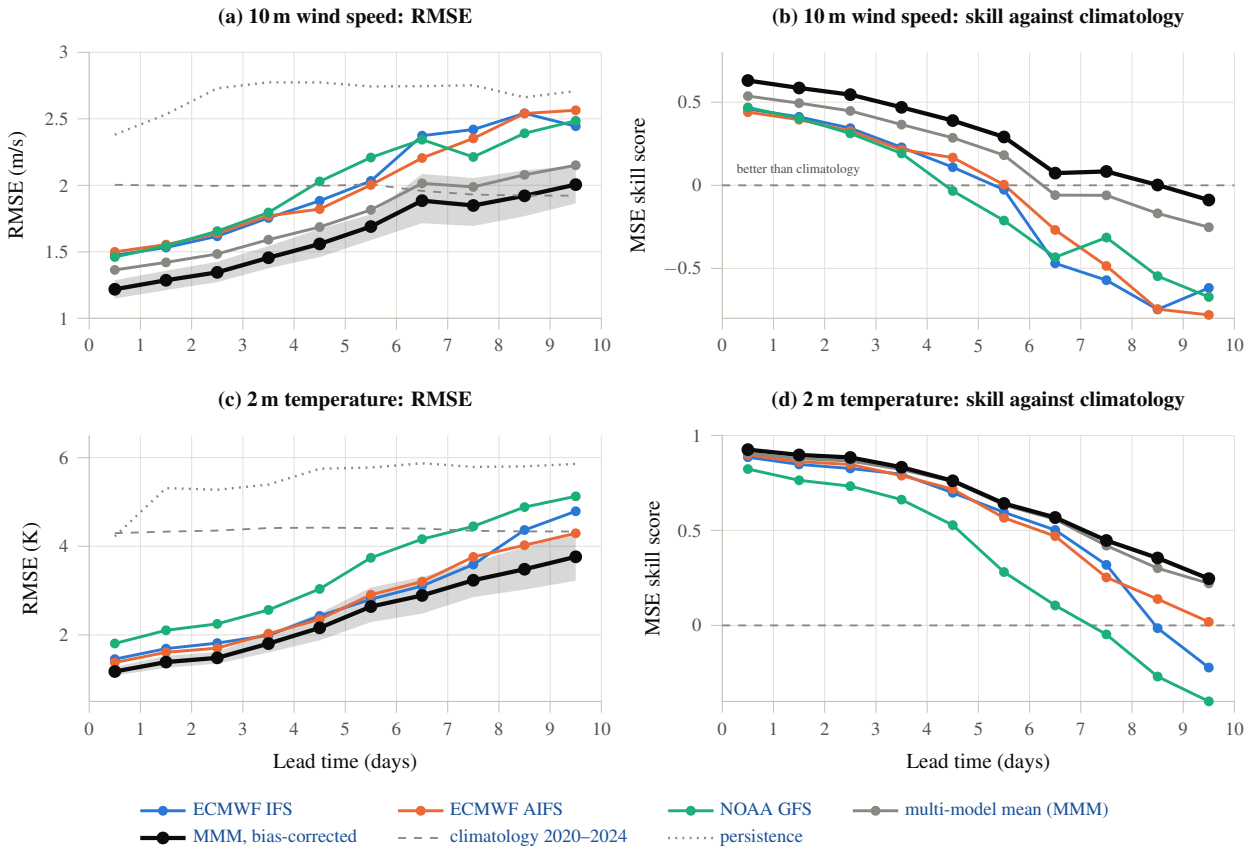


Figure 5: Skill decays within days; averaging and bias correction buy back about three. Verification at 12 UTC valid times for 61 runs (June–July 2025) at 62 German stations. MMM: multi-model mean of IFS, AIFS and GFS; BC: running 30-day bias correction. The grey band is the 95 % bootstrap interval of the bias-corrected MMM. The skill score is $1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{climatology}}$ with the 2020–2024 station climatology; below zero the forecast is worse than climatology.

Amplification (Proposition 1). Where both forecast and observation lie in the partial-load range, the ratio of logarithmic power error to logarithmic wind error has a median of 3.18 (interquartile range 3.05–3.41, 4,588 pairs), and 80 % of pairs reach the bound of 3 (Figure 7b); the remainder lie just below rated speed, where the real power curve bends over and the cubic idealisation no longer holds. Over all conditions, including calm hours below cut-in, the relative MAE of power is 1.9 times that of wind speed (57 % vs. 30 % at day-ahead), and the day-ahead MAE is 8.2 % of capacity at a mean capacity factor of 16 %.

Bidding (Proposition 4). Day-ahead quantile forecasts of pseudo power were built from conditional empirical quantiles: for each class of forecast power, the June runs (5,345 cases) give the quantiles of the observed power, which are then bid on the July runs (5,627 cases). With a shortfall three times as costly as a surplus, Proposition 4 predicts the optimal quantile $\tau^* = c_l/(c_l + c_s) = 0.25$. The test cost is lowest near $\tau = 0.200$, the cost at τ^* is within 0.8 % of that minimum, and bidding the τ^* quantile instead of the point forecast lowers the expected cost by 25 % (Figure 7c). With symmetric costs the median and the point forecast perform almost identically, as they should. The asymmetry of real imbalance prices is therefore a direct, measurable reason to produce probabilistic forecasts.

11 Visualising forecasts

The right chart depends on who reads it and which decision follows (Table 8). Four rules apply everywhere. Show uncertainty, as quantile bands or ensemble members, rather than a single line. State the time zone and interval convention and mark the present and the relevant gate closures. Use one categorical colour per entity throughout a report, never two y-axes in one chart, and dashed lines only for references and thresholds. Publish the numbers behind every chart, so that the chart is never the only way to read a value.

The figures in this paper follow these rules with a colour palette validated for colour-vision deficiency: neighbouring categorical colours are separated by at least 9 units of ΔE (OKLab$\times$100) under simulated protanopia and deuteranopia and by at least 19.6 for normal vision, magnitudes use a single-hue blue ramp, and every figure's data are in `data/processed`.

Table 7: Day-ahead verification against 62 German stations (lead times 24, 36 and 48 h; 61 runs from 1 June to 31 July 2025). Bias is forecast minus observation. BC: running 30-day bias correction using past data only. Intervals are 95 % bootstrap intervals over runs. Data: `data/processed/tab07_day_ahead_summary.csv`.

	10 m wind speed (m/s)			2 m temperature (K)		
Forecast	Bias	MAE	RMSE [95 % CI]	Bias	MAE	RMSE [95 % CI]
ECMWF IFS	0.02	1.06	1.37 [1.32, 1.42]	0.08	1.19	1.56 [1.49, 1.64]
ECMWF AIFS	−0.07	1.06	1.36 [1.31, 1.41]	0.32	1.15	1.51 [1.43, 1.59]
NOAA GFS	0.24	1.14	1.46 [1.40, 1.52]	−0.67	1.62	2.05 [1.98, 2.13]
Multi-model mean	0.06	1.02	1.30 [1.25, 1.35]	−0.09	1.12	1.46 [1.40, 1.51]
ECMWF IFS (BC)	−0.03	0.91	1.20 [1.15, 1.24]	−0.01	1.15	1.51 [1.45, 1.58]
ECMWF AIFS (BC)	−0.05	0.87	1.14 [1.10, 1.19]	−0.04	1.04	1.35 [1.30, 1.40]
NOAA GFS (BC)	−0.03	0.98	1.28 [1.22, 1.34]	0.17	1.34	1.75 [1.68, 1.83]
Multi-model mean (BC)	−0.04	0.86	1.12 [1.08, 1.16]	0.04	1.03	1.35 [1.30, 1.41]
Persistence	−0.03	1.62	2.16	0.12	3.17	4.16
Climatology 2020–2024	−0.24	1.25	1.66	−0.37	2.67	3.47

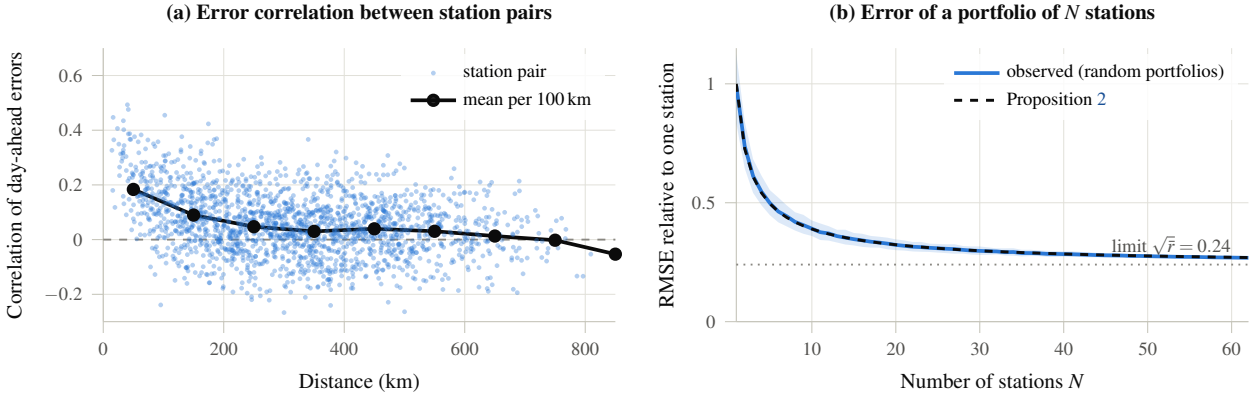


Figure 6: Local errors average out; shared errors set a floor. Day-ahead errors of 10 m wind speed of the bias-corrected multi-model mean, June–July 2025. (a) Correlation of errors for all 1,891 station pairs against distance, with the mean per 100 km bin. (b) RMSE of the average error of N randomly chosen stations relative to the mean single-station RMSE (line: mean of 300 draws; band: 10th–90th percentile), and the prediction of Proposition 2 with the observed mean correlation $\bar{r} = 0.06$.

12 Limitations

The requirements in Sections 3 to 5 summarise practice and literature; they are an expert synthesis, not a survey, and their thresholds (for example crane wind limits) must always be taken from the equipment and permit at hand. Market rules and data licences change; the facts quoted carry access dates and sources so that they can be rechecked.

The case study covers one summer and verifies 10 m wind and 2 m temperature at airports, not hub-height wind, irradiance or plant output: no openly accessible archive of measured irradiance and plant power for 2025 was part of the data set, and station winds include representativeness errors that a 0.25° model cannot resolve, which is one reason the raw biases are large. The pseudo power of Section 10 isolates the power-curve effect but ignores wakes, availability, the real wind profile and density. The bias correction is deliberately simple, and established post-processing (EMOS, quantile regression, gradient boosting with many predictors) would likely add more. Winter, with stronger winds, frontal systems, low stratus and snow, would test the models differently, and the comparison between IFS and AIFS reflects the operational versions of 2025 only.

13 Conclusion

A weather forecast for renewable energy is good only relative to a decision. Fixing the technology, the department and the horizon determines which variables matter, at what resolution and by when, in which uncertainty format, and with which scoring rule. A PV trader needs quarter-hourly irradiance quantiles before noon; a crane supervisor needs gusts at hub height for the next morning; a bank needs the P90 of twenty years. Four results govern the design of every such system: the power curve amplifies wind errors by a factor of at least three at partial load; errors shared across sites cannot be di-

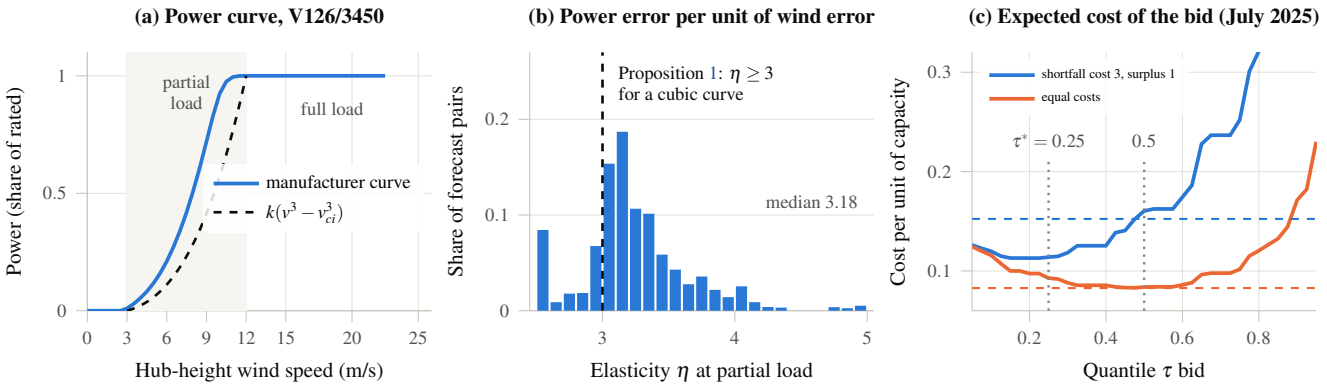


Figure 7: The power curve amplifies wind errors, and asymmetric costs call for a quantile bid. Pseudo power from observed and forecast 10 m winds scaled to 120 m (bias-corrected multi-model mean, day-ahead). (a) The turbine's power curve and its cubic idealisation between cut-in and rated speed. (b) Ratio of logarithmic power error to logarithmic wind-speed error for 4,588 pairs in which forecast and observation are both at partial load; values outside 2.5–5 are counted in the end bins. (c) Expected test cost of bidding the τ-quantile of conditional quantile forecasts (solid) against bidding the point forecast (dashed); the dotted lines mark the optima predicted by Proposition 4.

Table 8: Charts that answer each audience's question.

Audience	Charts
Trading	fan chart of quantiles with actuals; run-to-run change of the forecast; ensemble plume; residual load with price forecast; heat map of quarter-hours by day
Operations	meteogram with activity limits (gusts, waves, lightning); warning maps with plant locations; satellite loops
Asset management	expected vs. actual; power vs. irradiance or wind scatter; carpet plot of day by time of day; loss waterfall; cumulative yield vs. budget with P50/P90 band
Development	resource maps; wind rose; Weibull histogram; month-by-hour profile; exceedance curve; interannual variability
Construction	weather-window calendar; monthly workability from hindcasts
Forecast analysts	error and skill by lead time; reliability diagram; rank and PIT histograms; error by hour and month; worst-days gallery

versified away; averaging two models helps only when they are similarly good and fail differently; and under asymmetric imbalance costs the right bid is a quantile, not the expected value.

The case study confirms all four on public data and adds two practical findings. A machine-learning model, ECMWF's AIFS, now matches its physics-based counterpart for day-ahead station wind and beats it for temperature, while NOAA GFS trails both. And the largest gains in the study came not from choosing a model but from what follows it: averaging three models and removing a running bias extended the useful daytime wind forecast from about five to about eight and a half days and cut the day-ahead error by about a sixth. The data, code and tables needed to repeat or extend the analysis are published with this paper; the obvious next steps are hub-height wind, measured irradiance and plant output, and a winter season.

A Reproducibility

The paper is a self-contained research compendium in the repository folder `research/01-weather-forecasting-for-renewable-energy`. Its structure:

```
paper/              main.tex, sections/, figures/,
                    tables/, generated/, references.bib
code/               config.py, common.py, 01_ to 07_
data/raw/           download manifests with SHA-256
data/cache/         large downloads (not versioned)
data/interim/       observations, forecast point values
data/processed/     one CSV per figure and table
data/literature/    quoted facts with sources
supplement/         extended handbook (Markdown)
```

Pipeline. `make fetch` downloads the inputs: (1) station observations and the 2020–2024 climatology; (2) forecast extraction for the three models (about 12 GB are read with byte-range requests, 5 MB of point values are kept). `make analysis` runs the rest: (3) matching of forecasts and observations; (4) skill by lead time, bias correction and Table 7; (5) spatial aggregation; (6) power amplification and bidding; (7) the PV chain. `make paper` builds the PDF. Every script states its inputs and outputs in its header, and all settings live in `code/config.py`.

Figures and tables. Each figure file reads its data with `pgfplots` from the CSV named in its header; each case-study number in the text is a macro defined in `paper/generated/`. Table 9 maps results to their data and code.

Software and data licences. Python 3.13 with pandas, NumPy, ecCodes, pvlib [27] and windpowerlib [57]; LaTeX with pgfplots. ECMWF open data are used under CC-BY-4.0 (© ECMWF); NOAA GFS and ISD data are in the public domain; the turbine power curve comes from the OpenEnergy

Table 9: Where each result comes from. Script numbers refer to `code/`.

Result	Data file in `data/processed/`	Script
Fig. 3	`fig03_pv_chain_clear_day.csv`	07
Fig. 5	`fig05_skill_by_lead_*.csv`	04
Fig. 6	`fig06a_*.csv`, `fig06b_*.csv`	05
Fig. 7	`fig07a_*`, `fig07b_*`, `fig07c_*.csv`	06
Table 7	`tab07_day_ahead_summary.csv`	04
Prop. 3	`combination_check.csv`	04
Stations	`stations.csv`	03
Tables 4–6	`data/literature/*.csv`	n/a

turbine library distributed with windpowerlib.

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